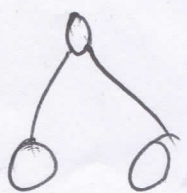
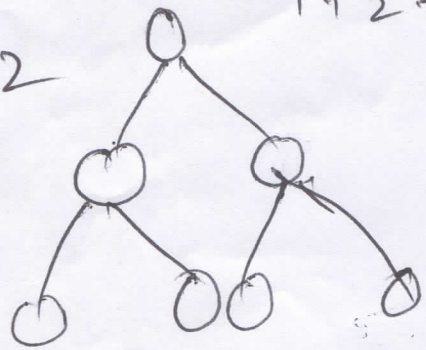
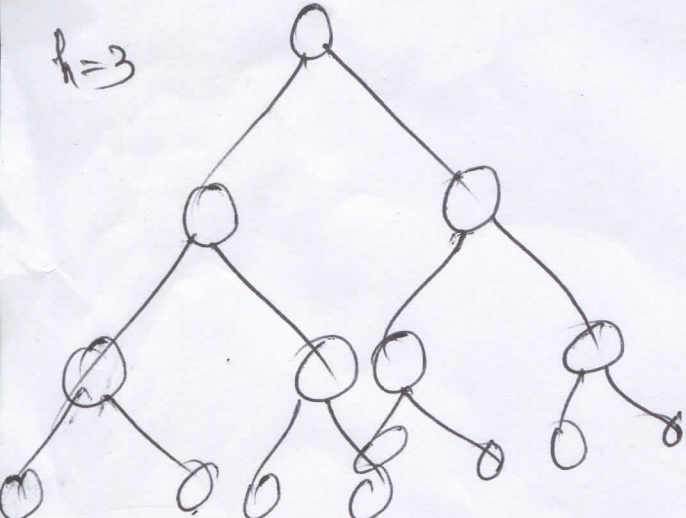


For ~~Tree~~ of height h

Max
 $h=0$ 0 # of nodes, 1

$h=1$  $1+2^1$

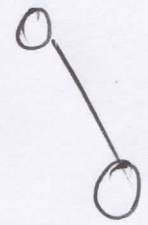
$h=2$  $1+2^1+2^2$

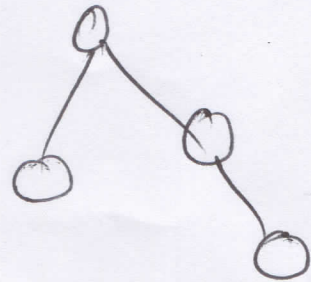
$h=3$ 

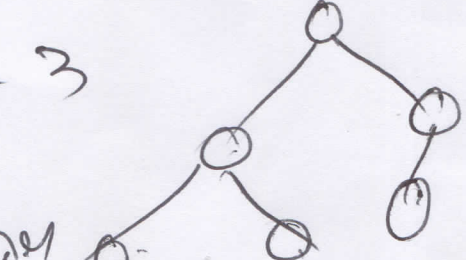
$$1 + 2^1 + 2^2 + 2^3 + \dots + 2^h$$

$$= 1 \frac{2^{h+1} - 1}{2 - 1} = 2^{h+1} - 1$$

Min # of nodes,
 $h=0$ 0 1

$h=1$  2

$h=2$  4

$h=3$  7

of nodes
 height of the tree

$$N(h) = 1 + n(h-1) + n(h-2)$$

for height 0 # of nodes
 $N(0)$ 1
 $N(1)$ 2
 $N(2)$ 4

$$N(3) = 1 + N(2) + N(1)$$

$$N(3) = 1 + 4 + 2 = 7$$

$$N(4) = 1 + N(4-1) + N(4-2) = 12$$

$$N(5) = 1 + N(5-1) + N(5-2) = 26$$

$$N(6) = 1 + N(6-1) + N(6-2) = 57$$